COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 007	**SECTION:**	CPE11S4
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ACTIVITY NO. 4.2 Characters and Strings

1. PROCEDURE OUTPUT

What procedures did you perform?
1. Try to create a program that outputs the following:
Testing islower:
p is a lowercase letter
P is not a lowercase letter
5 is not a lowercase letter
! is not a lowercase letter

Testing isupper:
D is an uppercase letter
d is not an uppercase letter
8 is not an uppercase letter
& is not an uppercase letter

u converted to uppercase is U
7 converted to uppercase is 7
$ converted to uppercase is $
L converted to lowercase is l

```cpp
#include <iostream>
#include <cstring>
#include <string>
#include <cctype>
using namespace std;

int main() {
// Testing is lower

char ch; //This is a character Constant
cout << "Enter four(4) character: \n";
cout << "Testing islower()\n";
for (int i=0;i<4;i++){
    cout << "Enter a character: ";
    cin >> ch;

    if(islower(ch)){
        cout << ch << " is a lowercase letter\n";
    } else {
        cout << ch << " is not a lowercase letter\n";
    }
}
cout << endl;
cout << "Testing isupper()\n";
for (int i=0;i<4;i++){
    cout << "Enter a character: ";
    cin >> ch;

    if(isupper(ch)){
        cout << ch << " is a uppercase letter\n";
    } else {
        cout << ch << " is not a uppercase letter\n";
    }
}
cout << endl;
cout << "Conversion\n";
for (int i=0;i<4;i++){
    cout << "Enter a character: ";
    cin >> ch;
```

```
}
cout << endl;
cout << "Conversion\n";
for (int i=0;i<4;i++){
    cout << "Enter a character: ";
    cin >> ch;

    if(islower(ch)){
        cout << ch << " converted to uppercase: " << (char)toupper(ch) << endl;
    }else if (isupper(ch)){
        cout << ch << " converted to lowercase: " << (char)tolower(ch) << endl;
    } else {
        cout << ch << " is not convertible. " << endl;
    }
}
return 0;
}
```

```
PS C:\CPE11S4-DOMINGO> & C:\CPE11S4-DOMINGO\friday.exe
Enter four(4) character:
Testing islower()
Enter a character: p
p is a lowercase letter
Enter a character: P
P is not a lowercase letter
Enter a character: 5
5 is not a lowercase letter
Enter a character: !
! is not a lowercase letter

Testing isupper()
Enter a character: D
D is a uppercase letter
Enter a character: d
d is not a uppercase letter
Enter a character: 8
8 is not a uppercase letter
Enter a character: &
& is not a uppercase letter

Conversion
Enter a character: u
u converted to uppercase: U
Enter a character: 7
7 is not convertible.
Enter a character: $
$ is not convertible.
Enter a character: L
L converted to lowercase: l
PS C:\CPE11S4-DOMINGO>
```

EXPLANATION: This program demonstrates how to check and convert characters using functions from the `cctype` library. First, it asks the user to enter four characters and uses `islower()` to determine whether each character is a lowercase letter. It then asks for another four characters and uses `isupper()` to check if they are uppercase letters. In the last part, the program converts lowercase characters into uppercase using `toupper()` and uppercase characters into lowercase using `tolower()`. If the entered character is not a letter, the program displays that it cannot be converted.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

For each of the following, Create the following C++ Code and output for each case:
1. Write a C++ program that accepts a character input from the keyboard and tests the character using each function in the <cctype> character-handling library. Display the result of each function.

```cpp
#include <iostream>
#include <cctype>
using namespace std;

int main() {
    char ch;

    cout << "Enter a character: ";
    cin >> ch;

    cout << "\nResults:\n";

    cout << "isalnum: " << isalnum(ch) << endl;
    cout << "isalpha: " << isalpha(ch) << endl;
    cout << "iscntrl: " << iscntrl(ch) << endl;
    cout << "isdigit: " << isdigit(ch) << endl;
    cout << "isgraph: " << isgraph(ch) << endl;
    cout << "islower: " << islower(ch) << endl;
    cout << "isprint: " << isprint(ch) << endl;
    cout << "ispunct: " << ispunct(ch) << endl;
    cout << "isspace: " << isspace(ch) << endl;
    cout << "isupper: " << isupper(ch) << endl;
    cout << "isxdigit: " << isxdigit(ch) << endl;

    cout << "tolower: " << char(tolower(ch)) << endl;
    cout << "toupper: " << char(toupper(ch)) << endl;

    return 0;
}
```

```
Problems  Output  Debug Console  Terminal  Ports

PS C:\CPE11S4-DOMINGO> & C:\CPE11S4-DOMINGO\testtig.exe
Enter a character: a

Results:
isalnum: 2
isalpha: 2
iscntrl: 0
isdigit: 0
isgraph: 2
islower: 2
isprint: 2
ispunct: 0
isspace: 0
isupper: 0
isxdigit: 128
tolower: a
toupper: A
PS C:\CPE11S4-DOMINGO>
```

EXPLANATION: This program demonstrates different character testing and conversion functions from the `cctype` library. It asks the user to enter one character and then checks different properties of that character using functions such as `isalnum()`, `isalpha()`, `isdigit()`, `islower()`, `isupper()`, and `isspace()`. It also uses `iscntrl()`, `isgraph()`,

`isprint()`, `ispunct()`, and `isxdigit()` to identify other characteristics of the entered character. The program then uses `tolower()` and `toupper()` to convert the character into lowercase and uppercase. The results are displayed one by one so the user can see which conditions are true for the entered character.

2. Write a C++ program that accepts four strings representing integers, converts them into integer values, calculates their sum, and displays the total.

hoa4.2.1.cpp > main()

```cpp
#include <iostream>
#include <cstring>
#include <string>
#include <cctype>
using namespace std;


int main () {

    string num1, num2, num3, num4;

    cout << "Enter a number: ";
    cin >> num1;
    cout << "Enter a number: ";
    cin >> num2;
    cout << "Enter a number: ";
    cin >> num3;
    cout << "Enter a number: ";
    cin >> num4;

    int n1 = stoi(num1), n2 = stoi(num2), n3 = stoi(num3), n4 = stoi(num4);
    cout << endl;
    cout << "Sum is: " << n1 + n2 + n3 + n4;


    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\CPE11S4-DOMINGO> & C:\CPE11S4-DOMINGO\hoa4.2.1.exe
Enter a number: 1
Enter a number: 6
Enter a number: 34
Enter a number: 8

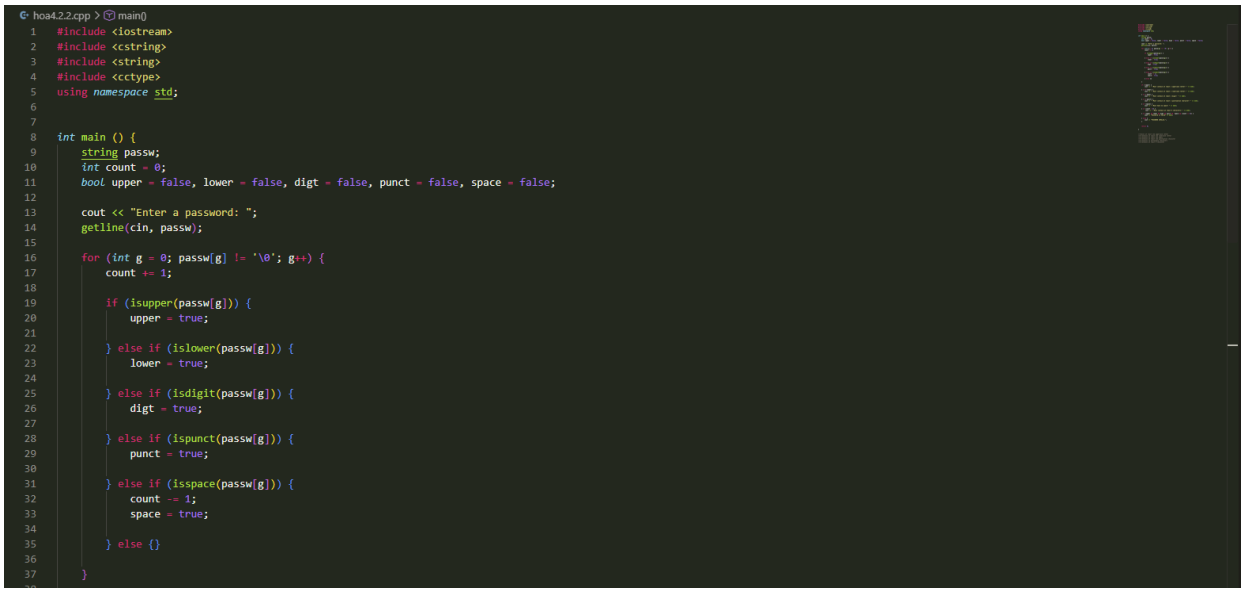
Sum is: 49
PS C:\CPE11S4-DOMINGO>
```

EXPLANATION: This program asks the user to enter four numbers and stores them first as strings. The `stoi()` function converts these string values into integers so they can be used in addition. After converting all four inputs, the program adds them together and displays the total as the sum. It shows how string values can be converted into integers before performing a basic mathematical operation.

3. Write a C++ program that accepts a password as a string and determines whether it meets all of the following requirements:

- Contains at least one uppercase letter
- Contains at least one lowercase letter
- Contains at least one digit
- Contains at least one punctuation character
- Contains no whitespace characters
- Contains at least 8 characters

The program must use appropriate functions from the <cctype> library to examine each character. Display **which requirements are satisfied or not satisfied,** and indicate whether the password is **VALID** or **INVALID**



```cpp
hoa4.2.2.cpp > main()
#include <iostream>
#include <cstring>
#include <string>
#include <cctype>
using namespace std;


int main () {
    string passw;
    int count = 0;
    bool upper = false, lower = false, digt = false, punct = false, space = false;

    cout << "Enter a password: ";
    getline(cin, passw);

    for (int g = 0; passw[g] != '\0'; g++) {
        count += 1;

        if (isupper(passw[g])) {
            upper = true;

        } else if (islower(passw[g])) {
            lower = true;

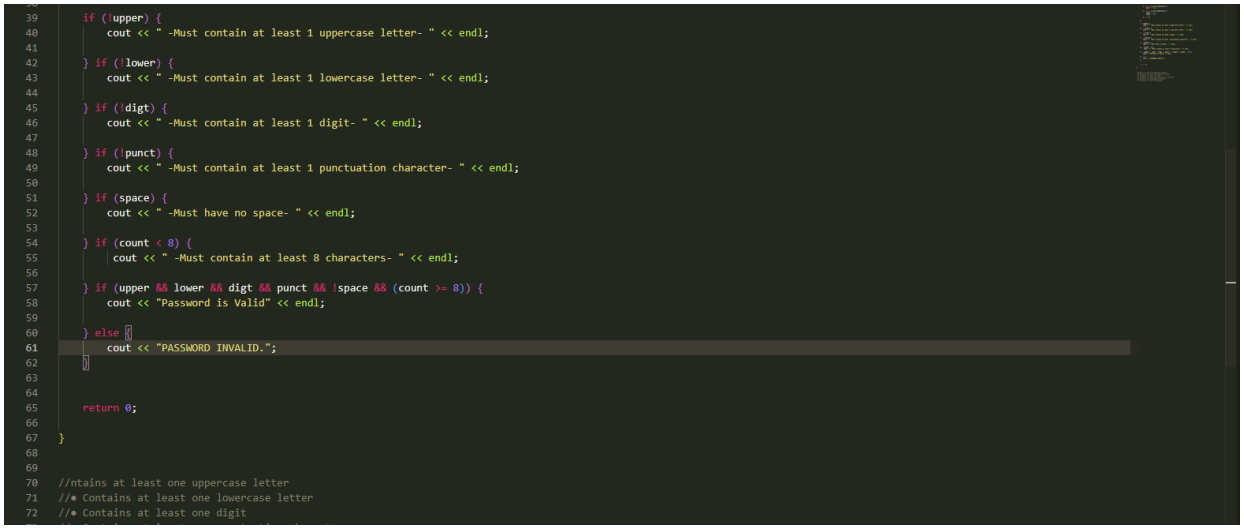
        } else if (isdigit(passw[g])) {
            digt = true;

        } else if (ispunct(passw[g])) {
            punct = true;

        } else if (isspace(passw[g])) {
            count -= 1;
            space = true;

        } else {}

    }
```



```cpp
    if (!upper) {
        cout << " -Must contain at least 1 uppercase letter- " << endl;

    } if (!lower) {
        cout << " -Must contain at least 1 lowercase letter- " << endl;

    } if (!digt) {
        cout << " -Must contain at least 1 digit- " << endl;

    } if (!punct) {
        cout << " -Must contain at least 1 punctuation character- " << endl;

    } if (space) {
        cout << " -Must have no space- " << endl;

    } if (count < 8) {
        cout << " -Must contain at least 8 characters- " << endl;

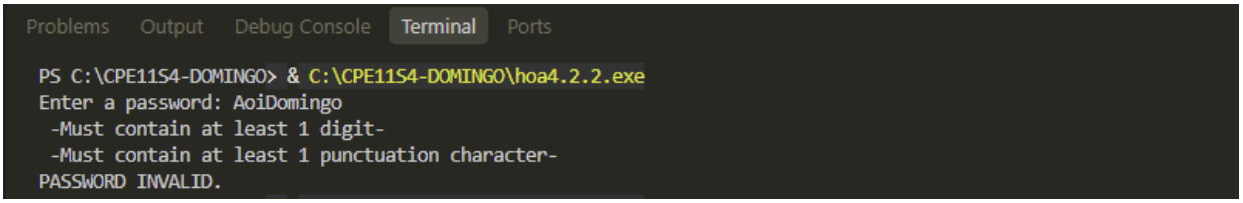
    } if (upper && lower && digt && punct && !space && (count >= 8)) {
        cout << "Password is Valid" << endl;

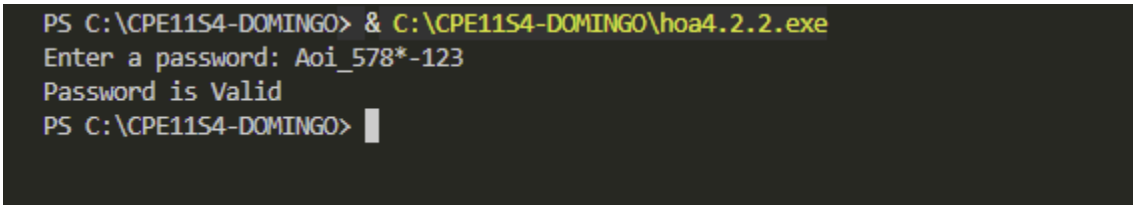
    } else {
        cout << "PASSWORD INVALID.";
    }


    return 0;

}


//ntains at least one uppercase letter
//• Contains at least one lowercase letter
//• Contains at least one digit
```





EXPLANATION: This program checks if a password follows several requirements before considering it valid. It uses `getline()` to get the whole password, including spaces, and then loops through each character to check for uppercase letters, lowercase letters, digits, punctuation, and spaces. The `count` variable keeps track of the password length while spaces are excluded from the character count. After checking all the characters, the program displays the requirements that were not met. If the password has at least 8 non-space characters, contains all the required character types, and has no spaces, it displays “Password is Valid”; otherwise, it displays “PASSWORD INVALID.”

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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